C^3: Committed Confidence Calibration Selection Bias in Confidence-Based Token Commitment in Diffusion Language Models

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Abstract

Diffusion language models commit several tokens per denoising step, and almost every published decoder chooses which by ranking the masked positions on predicted probability and keeping the highest scores. Ranking on an estimate and keeping the winners is a selection rule, and selection rules favour candidates whose estimation error happened to run high, so the confidence attached to a committed token may overstate the model's actual certainty. We propose to measure whether it does, by asking whether reported confidence remains a sufficient statistic for correctness once a token's rank and the size of the field it beat are known. The measurement is made against held-out ground truth under teacher-forced infilling, which avoids the dependency confound a sequential-decoder reference would introduce, and identification comes from randomising the composition of the candidate pool at a fixed model state. We then use the same instrument to separate the two additive terms in parallel-decoding error, the dependency term and the marginal estimation term, which existing evaluations report as entangled: exactly and in nats on a synthetic probe where the data distribution is known by construction, and as identified contrasts rather than levels on real corpora.

1 Introduction

Diffusion language models (dLLMs) generate text by repeatedly denoising a masked sequence (Austin et al., 2021a; Sahoo et al., 2024). The model predicts a distribution at every masked position in one forward pass, so a single step can commit several tokens. Open checkpoints such as LLaDA-8B (Nie et al., 2025) and Dream-7B (Ye et al., 2025) have made the approach practical, and the operational question is how many positions a step can safely fill.

Published decoders answer it similarly. Each step yields, for every masked position, a candidate token and the probability the model assigns to it, conventionally called the position's confidence. The decoder commits the positions holding the highest values (Chang et al., 2022; Nie et al., 2025; Wu et al., 2025).

Confidence is an estimate produced by a model of finite capacity, and estimates carry error. Because the decoder keeps whichever positions scored highest, the maximum over several noisy estimates will tend to belong to a candidate whose error ran in the favourable direction. Economists and geneticists call the resulting upward bias the winner's curse (Andrews et al., 2024; Efron, 2011). Averaged over commitments, it predicts that the number attached to a committed token overstates the model's certainty by an amount that grows with the size of the field the token beat, and that field is largest early in a generation and for long outputs.

We test this as a claim about sufficiency: if confidence is calibrated given selection, then knowing a token's rank and field size adds nothing to what its confidence already says about whether it is correct.

A second explanation for why parallelism costs quality already exists: the model supplies marginals and never represents the joint, so tokens committed together can be individually plausible and jointly incoherent. The two mechanisms are additive. For a committed set S with context x_{obs},

$$\mathrm{KL}\Big(p^* \,\Big\|\, \textstyle\prod_{i\in S} p_{\theta,i}\Big) = \underbrace{\mathrm{TC}(x_S \mid x_{\text{obs}})}_{\text{(A) dependency}} + \underbrace{\textstyle\sum_{i\in S} \mathrm{KL}\big(p_i^* \,\big\|\, p_{\theta,i}\big)}_{\text{(B) estimation}} \quad (1)$$

Equation 1 is the familiar decomposition of a KL divergence to a product distribution, used here for framing and not claimed as theory. The dependency literature targets (A) and treats (B) as a constant. Selecting on a noisy estimate inflates (B), and one observable separates the two: (A) is indifferent to which position ranked first, so no dependency account predicts a gradient in accuracy across ranks within a step.

2 Motivation

Three shortcomings in existing work point to this direction.

The commit statistic is tuned but never audited. Decoders in the confidence-threshold family select a cutoff on the confidence values and report the resulting speed and quality (Wu et al., 2025; Shen and Ro, 2025) without reporting whether those values are calibrated. A threshold is a statement about a quantity whose meaning has not been established, so one tuned on a given benchmark has no interpretation elsewhere.

Two error sources are measured as one. Tang et al. (2026) observe that in current evaluations of discrete diffusion samplers, sampler error remains entangled with model approximation error, and do not separate them. The dependency literature (B. Kim et al., 2026; Ringel et al., 2026; Kang et al., 2026) attributes degradation entirely to the joint-versus-marginal gap and assumes the marginals are accurate. If part of it comes instead from the marginals being miscalibrated at the states selection visits, effort is going to the wrong term.

Corrections rest on unmeasured premises. PC-Sampler calibrates the commit statistic with a position-aware weight and an adjustment suppressing trivial tokens (Huang et al., 2025); KLASS and SWD reweight selection using inter-step stability (S. H. Kim et al., 2025; Wu and Huang, 2026). Each corrects a quantity whose failure mode is uncharacterised. Knowing whether the miscalibration tracks rank and field size, and how large it is, would say which corrections should work and why.

3 Background and Related Work

Confidence- and stability-based selection. Ranking by predicted probability originates with MaskGIT (Chang et al., 2022) and is the LLaDA and Dream default; the statistic varies, with margin (J. Kim et al., 2025) and entropy (Ben-Hamu et al., 2025) in use, as does the cutoff, from one global threshold in Fast-dLLM (Wu et al., 2025) to OSDT's schedule calibrated on a single sequence and reused, justified by confidence trajectories being nearly identical across inputs from one dataset (Shen and Ro, 2025). A second group ranks on stability: KLASS uses token-level KL divergence across steps (S. H. Kim et al., 2025), SWD proves this instability is a strict lower bound on a token's mutual information with the remaining masked context and uses it to reweight any selector (Wu and Huang, 2026), LESS adds persistence (Mohamed et al., 2026) and STDec spatial structure (Y. Chen et al., 2026). SWD's theorem constrains any reading of inter-step movement as estimation error, since the raw signal carries dependency information.

Calibration of the commit statistic. PC-Sampler is the closest existing work to a correction of the kind our measurement would motivate (Huang et al., 2025): it identifies a bias toward trivial tokens early in decoding and corrects it with a position-aware weighting and a calibrated confidence score. That bias is positional and content-driven. Ours is a property of the ranking operation, so it should appear even among positions of equal difficulty and position, and should scale with field size. The predictions are separable, and we enter PC-Sampler's weight into our own specification as a competing covariate (§7).

Dependency-aware decoding. DAPD commits independent sets recovered from self-attention (B. Kim et al., 2026); DEMASK learns pairwise conditional influence with a total-variation bound (Ringel et al., 2026); Bansal and Sanghavi (2025) add an auxiliary sampler for approximate joint sampling; PUNT selects by conditional independence testing (Azangulov et al., 2026); APD uses an autoregressive verifier (Israel et al., 2025); ParallelBench proves subadditivity of conditional total correlation (Kang et al., 2026). All of it targets term (A).

Work	Calibration under selection	Separates (A) / (B)
Fast-dLLM; OSDT	no	no
KLASS; SWD; LESS	no	no
PC-Sampler	pos. bias only	no
DAPD; DEMASK; PUNT	no	(A) only
ParallelBench	no	(A) only
Jazbec et al. (2026)	no	no
Cai and Li (2026)	no	no
Tang et al. (2026)	no	identifies
This work (proposed)	yes	both

Table 1: Position relative to the closest existing work. The final row states what we intend to deliver, not an established result.

Learning the statistic away. Jazbec et al. (2026) train unmasking policies with reinforcement learning, and Z. Chen et al. (2026) distil a checkpoint into reaching high certainty faster; both replace the hand-specified statistic rather than auditing it. This is the strongest challenge to our premise, so we state the answer plainly. A learned policy can absorb a selection bias without exposing it, and therefore cannot say whether the bias exists, how large it is, or what it tracks; and the training-free decoders that inherit those questions remain the default in every open checkpoint we can run.

Theory and evaluation. Cai and Li (2026) bound the iteration complexity of an entropy-sum decoder, which unmasks until cumulative entropy crosses a threshold, taking the predicted distributions as given. Frkovic et al. (2026) find a representative post-hoc remasking method performs on par with confidence unmasking in the standard short-block greedy setting. Table 1 places this work: we are aware of no measurement of calibration conditional on the selection event, and none of the two terms in Eq. 1 estimated separately.

4 Contributions

We claim two primary contributions and one contingent on their outcome.

C1. A selection-conditional calibration audit. We measure whether reported confidence stays sufficient for correctness once rank and field size are known, against held-out ground truth rather than a model-derived reference, with rank and field size randomised by design. §5.1 shows the question is empty unless approximation error varies in scale across positions, so C1 also delivers

the first estimate of that dispersion, which is the quantity governing whether any selection correction can help. This addresses the first shortcoming in §2, converting the threshold from a tuned number into a quantity with a characterised failure mode. We release the traces so the audit can be repeated without repeating the compute.

C2. Separate estimates of the two error terms. On the synthetic probe of §6 the data distribution is known, so both terms of Eq. 1 are computed exactly in nats and the parallel-versus-sequential contrast can be calibrated against the true (A). On real corpora held-out tokens identify the cross-entropy $\mathrm{CE}(p_i^*, p_{\theta,i})$ but not $H(p_i^*)$, so (B) is identified only up to an additive constant. That constant is fixed within a masked state, so contrasts in (B) across field sizes and mask distributions are identified even though levels are not, and contrasts are what we report; likewise the disagreement rate is a probability rather than a divergence, so on real corpora we report it as a monotone proxy for (A), calibrated on the probe. Reporting both against tokens committed per step addresses the entanglement Tang et al. (2026) identify and indicates which term binds in the regime these decoders are run at.

C3, contingent. A zero-cost correction. If C1 finds a field-size or rank effect predictable from an inter-step signal, the same regression yields a shrinkage rule costing no extra forward passes. We attempt C3 only if C1 succeeds and report the null otherwise, since it would corroborate SWD's account of what inter-step movement measures.

Risk-controlled thresholding, which would turn a tuned cutoff into a user-specified tolerance with finite-sample validity (Angelopoulos et al., 2024, 2025), is deferred. It needs a calibration budget beyond this project's scope, and the closest prior work (Schuster et al., 2022; Jazbec et al., 2024) covers stopping and acceptance decisions rather than commitment, so it remains open.

5 Methodology

5.1 Setup and the Selection Effect

Let m be the size of a candidate pool drawn from masked set M, v_i the model's top-1 token at position i, and r its rank. We write the log-odds of the model's confidence as $\ell_i = \ell_i^* + \varepsilon_i$, where ℓ_i^* is the true log-odds and ε_i is position-specific approximation error. Retaining the maximum of noisy estimates naturally biases ℓ upward.

Crucially, conditioning on reported confidence ℓ_i makes the selection event ancillary if the noise scale is uniform across positions: the null and alternative hypotheses make the exact same prediction. Field size m becomes informative given ℓ_i exactly when noise scales vary across positions (heteroskedasticity), as extreme confidence values disproportionately originate from high-variance positions when m is large. We test this operationally relevant hypothesis: whether the exact same reported confidence means different things depending on the field size it beat.

5.2 Measurement Instrument and Identification

A sequentialised reference confounds selection bias with local dependency structure. We instead use held-out ground truth. Running one forward pass on a true document x^0 with masked positions yields ℓ_i, v_i, and the true token x_i^0, letting us observe correctness directly as $\mathrm{agree}_i = \mathbf{1}\{v_i = x_i^0\}$ free of joint commitment confounders. We evaluate under uniform masks (matching training) and decoder-induced masks (simulating the inference trajectory) to measure selection-induced covariate shift.

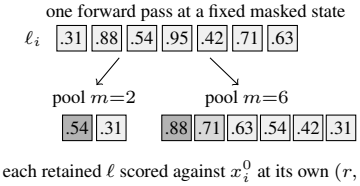


Figure 1: At a fixed state, random subsets $P \subset M$ of varying sizes m are drawn. Only pool composition varies, making selection intensity exogenous given reported confidence.

For identification, we draw random subsets of varying sizes $m \in \{2, 4, 8, 16, 32\}$ at a fixed state, retaining the top positions. Since one forward pass is shared across draws, the model state is strictly constant and difficulty is balanced. Selection intensity becomes strictly exogenous given reported confidence (Figure 1). We evaluate this via two regressions with state fixed effects and document-clustered standard errors:

$$\Pr(\text{agree}) = \sigma(\alpha_{\text{state}} + f(\ell_i) + \beta_z z_{r,m}) \tag{2}$$

$$\Pr(\text{agree}) = \sigma\Big(\alpha_{\text{state}} + f(\ell_i) + \textstyle\sum_{k\geq 2} \gamma_k \mathbf{1}\{r = k\}\Big) \tag{3}$$

Here, $f(\ell_i)$ is a flexible non-linear control for reported confidence to prevent linear terms from erroneously absorbing the confidence–accuracy curvature, and $z_{r,m}$ is the expected order statistic. In Eq. 2, calibration implies $\beta_z = 0$ and a winner's curse implies $\beta_z < 0$. In Eq. 3, any monotone rank profile ($\gamma_k \neq 0$) uniquely flags selection bias, as token dependency is indifferent to rank.

5.3 Stage 0: A Pilot with a Written Gate

Two parameters dictate if C1 is empirically feasible: the overall scale of the bias, and the dispersion of position-specific noise (without which $\beta_z = 0$). A 20-document pilot estimates these parameters to compute the sample size needed for 80% power. We pre-register a strict gate: proceed with the full study only if the required size is $\leq 1{,}000$ documents and the noise dispersion is sufficiently large. Otherwise, we restrict C1 to the 110M model where sampling is cheap, and shift computational weight entirely to C2.

6 Dataset

Held-out text. 2,000 documents sampled by fixed seed from the FineWeb-Edu 10BT subset (Penedo et al., 2024), truncated to 256 tokens and excluded from the 110M model's training set. This is the primary calibration corpus, since it supplies ground truth at every position.

Contamination. "Held-out" is guaranteed only for the 110M model we train ourselves. LLaDA-8B and Dream-7B were pretrained on web-derived corpora that are not fully disclosed, so these documents may have been seen. Memorised text would show inflated agreement at every rank and would bias the estimated *level* of calibration error, though not obviously its gradient in field size. We screen by prompting each checkpoint with a 50-token prefix, measuring verbatim continuation length, flagging documents above a fixed threshold, and reporting every endpoint with and without them. Claims about levels rest on the 110M arm; the 8B arm carries claims about the gradient only.

Reasoning and code. GSM8K, 1,319 test problems (Cobbe et al., 2021), and MBPP, the 500-problem test split (Austin et al., 2021b); reference and canonical solutions supply the ground-truth tokens for teacher-forced infilling. HumanEval, at 164 problems (M. Chen et al., 2021), is used only for end-task evaluation, being too small to support the fitting splits below.

Structured output. A controlled probe of 1,000 JSON instances generated from 200 schemas drawn from SchemaStore,[1] with values sampled independently from a fixed vocabulary under a generative process we specify. Because p^* is then known in closed form at the token level, both terms of Eq. 1 are computable exactly in nats here. This probe is therefore the anchor for C2 rather than an additional benchmark.

Splits and tokenisation. Fitting uses a 50/50 document-level split fixed by seed: coefficients in Eq. 2 and Eq. 3 are fitted on the first half and all reported estimates come from the second. Each model is evaluated under its own tokeniser, since LLaDA and Dream ship their own and the 110M model uses GPT-2 BPE, and documents are truncated to their first 256 tokens under that tokeniser and never packed, so every position lies inside a single document. Token counts per document therefore differ slightly across models, which the state fixed effects absorb. No other preprocessing is applied.

7 Evaluation Plan

Endpoints. The primary endpoint is β_z in Eq. 2, with a 95% interval from a document-clustered bootstrap over 10,000 resamples; we declare an effect if the interval excludes zero and the estimate exceeds twice the trace-level non-determinism floor, measured by re-running 20 documents five times. Co-primary is the rank profile γ_k of Eq. 3, which carries the discrimination against dependency accounts. Secondary: β_s raw and disattenuated; the dispersion of $\hat{s}_i$ from Stage 0 read against β_z, since a nonzero β_z under near-homoskedastic $\hat{s}_i$ would indicate residual misspecification rather than selection; the uniform-versus-decoder-induced contrast at matched mask rate; and the two term estimates against tokens committed per step.

Baselines, and what each is a baseline for. Because C1's endpoint is a coefficient rather than a decoder score, only two comparisons bear on it. A temperature tuned per step and per field size is the obvious competing explanation for any effect we report; a position's confidence depends on its whole logit vector, so temperature can reorder positions and the comparison has to be empirical, and temperature in dLLM sampling has itself been studied (Olausson et al., 2026). PC-Sampler's position-aware weight (Huang et al., 2025) enters Eq. 2 as a competing covariate: if adding it drives β_z to zero, the effect is positional rather than selection-induced. The decoder suite, namely confidence top-k, Fast-dLLM thresholding (Wu et al., 2025), entropy-bounded unmasking (Ben-Hamu et al., 2025), margin selection (J. Kim et al., 2025) and KLASS (S. H. Kim et al., 2025), enters only if C3 runs, where the comparison is task accuracy at matched tokens per step. If C1 returns a null, C3 and this suite are dropped together.

Models and precision. LLaDA-8B-Instruct (Nie et al., 2025) and Dream-7B (Ye et al., 2025) check that findings hold at deployed scale. One 110M model trained from scratch is the controlled instrument, where exact likelihoods are affordable and the training set is known; it follows the MDLM recipe of Sahoo et al. (2024) on the FineWeb-Edu 10BT subset with the calibration documents removed beforehand. Quantisation perturbs exactly the logits under audit, so a 16-bit replication on the small model licenses nothing about the large ones: we run the 8B models in 4-bit for the full sweep and re-run a 200-document subsample of LLaDA-8B in bfloat16 with CPU offload, reporting β_z under both precisions. If β_z is null we report it, since a null would explain why plain thresholding works as well as it does, and C2 does not depend on its sign.

8 Scope, Timeline and Compute

A second small model and risk-controlled thresholding were cut from an earlier draft, the first because it supported no measurement the 110M model does not, the second because its calibration budget cannot be funded alongside C1 and C2. Weeks 1–2 cover the pilot and gate, 3–4 the small model and trace schema, 5–7 the C1 audit including the bfloat16 and contamination controls, 8 the C2 estimates on the probe and on real corpora, 9 C3 and the decoder suite if the gate was passed, 10 ablations and write-up.

[1] https://github.com/SchemaStore/schemastore

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